

Liquid Metal's Elasticity

An experiment that compares the elasticity of a ball bearing made out of liquid metal with steel and titanium <http://dgit.in/NNX01X>



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HDTVs: IN BLACK AND WHITE

It is not always black and white with HDTVs. Here are five things you should look out for

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It is the so-called holy trinity – LEDs, Plasmas and LCDs. But is it? With most other devices, you have a fair idea of what the performance might be, after one look at the spec sheet. However, with a television, you really have no generalization in terms of specifications and hardware. While technically you may compare between two LED televisions, the resulting performance may be vastly different. As a consumer, this can become quite inundating on the shop floor, but you need to keep a few things in mind before splurging your cash on the most expensive HDTV.

LED and LCD – Are they really that different?

The answer is a resounding NO! In a nutshell, LED TVs are essentially LCD TVs, but with LED backlighting instead of the fluorescent backlighting in LCDs. Even now, there are two types of LED TVs – with EDGE backlighting or the Full Array (also known as Full LED). With the former, what you get is a line of LEDs placed on the outside edge of the panel, allowing the TVs to be slimmer than ever. You can easily identify this

kind of a TV by looking at it when it is switched on and not connected to any AV source. You will see “spotlighting” around the edges of the panel. While this TV may be slim, the black levels tend to be slightly less deep than full LEDs and the edges seem brighter than the centre of the display.

With Full Array, a bunch of LEDs are placed behind the screen, in multiple rows, leading to more even lighting across the panel. Another advantage is that manufacturers can deploy local dimming for individual LEDs, leading to deeper details in scenes. Even Edge LEDs are now using the local dimming feature, but using diffusers and guides to attempt at having a similar effect as actual dimming. From our experience, local dimming works well only on full LEDs, while the attempt on edge LED televisions mostly leads to uneven brightness, usually known as the halo effect - you see block of leaking light around a brightly lit object in the midst of black colour.

Contrast Ratios – Not always black and white

Let us start with two examples. The Bravia HX750 series has an actual contrast ratio of 1129:1, while the HX925 has a contrast ratio of 963:1. To put this in perspective, the HX925 is the most expensive HDTV series on the entire Bravia range

from Sony, while the HX750 is two notches below that. To put it simply, the adage that the more expensive the TV, the better the performance isn't necessarily true. And equally, take the contrast ratio numbers thrown around by TV manufacturers with a pinch of salt. We have a table of some of the TVs that we have tested (40/42-inches and above), with the actual contrast ratios, as tested. Incidentally, the most inexpensive television range in this entire list has the best contrast ratio of all. Let us specify that we use the Spyder 3 Elite benchmark, a colorimeter tool along with its software, to derive the actual scores. Along the way, we ensure that the contrast, brightness, backlight, advanced contrast and noise reduction settings are all equalized across all televisions to level the playing field. All televisions are tested in the exact same environment – lighting, room layout etc., so that all factors remain the same. Scores between 900 and 1200 and above in our benchmark tests indicate very good performance, with deep blacks.

Motion control – not the best attempt at making TV a physical activity!

This has the demographic completely divided into two camps. Let us look at the very basic tasks first – changing